

Total No. of Questions : 8]

[Total No. of Printed Pages : 2

Roll No

CE-3002-CBGS

B.E., III Semester

Examination, June 2020

Choice Based Grading System (CBGS)

Fluid Mechanics

Time : Three Hours

Maximum Marks : 70

- Note:** i) Attempt any five questions.
ii) All questions carry equal marks.

1. a) Explain the terms Specific volume, specific gravity, surface tension and viscosity. 7
b) Explain with neat sketches the conditions of equilibrium for floating and submerged bodies. 7
2. Deduce an expression for Bernoulli's equation and also mention its applications. 14
3. a) Explain the concept of water hammer transmission of power. 7
b) Explain the boundary layer theory. 7
4. a) Explain the terms Stream lines, streak lines and path lines. 7
b) Explain various types of fluid flow. 7
5. a) Explain the surges in open channels. 7
b) Explain the concept of channel flow routing. 7

CE-3002-CBGS

PTO

[2]

6. a) Differentiate between Gradually varied flow and Rapidly varied flow. 7
b) A trapezoidal channel having the side slope of 2H:3V carries water at $0.5\text{m}^3/\text{sec}$. If the bed slope of the channel is 1:1000, determine the optimum dimensions of the channel. Take Chezy's constant 'C' = 50. 7
7. a) Explain the different types of drags in detail. 7
b) Explain the construction and working of a pelton wheel with a neat sketch. 7
8. Write short notes on any four of the following: 14
 - a) Specific energy
 - b) Bulk modulus of elasticity
 - c) Vorticity and circulation
 - d) Losses due to sudden expansion and contraction
 - e) Draft tube theory
